SIDEBAR **Why GPyTorch**

There are other GP modeling library in Python, but we choose GPyTorch for this task for this book. Built on top of PyTorch, GPyTorch offers a streamlined workflow from array manipulation to GP modeling and eventually to Bayesian optimization with BoTorch, which we will start using in Chapter 4.

The library is also being actively maintained and has many state-of-the-art methods implemented. For example, Chapter 12 covers using GPyTorch to scale a GP to large data sets, and in Chapter 13 we learn to integrate a neural network into a GP model.

2.1 How to sell your house the Bayesian way

Before we jump right into GPs, let's consider an example scenario in the domain of housing price modeling, and how the price of a house is determined in relation with other houses. This discussion serves as an example of how correlation works in a multivariate Gaussian distribution, which is a central part of a Gaussian process.

Say we are a home owner in Missouri who is looking to sell off our house. We are trying a set an appropriate asking price, and are talking to a friend about how to do this.

Us: I'm not sure what to do. I just don't know exactly how much my house is worth.

Friend: Do you have a rough estimate?

Us: Somewhere between 150k and 300k would be my guess.

Friend: That's a pretty big range.

Us: Yeah, I wish I knew people who have sold their houses. I need some references.

Friend: I heard Alice sold her house for 250k.

Us: Alix who's in California? That's really surprising! Also, I don't think a house in California will help me make a better estimate for my own house. It could still be anything between 150k and 300k.

Friend: No, it's Alice who lives right next to you.

Us: Oh, I see. That's very useful actually, since her house is really similar to mine! Now I would guess that my house will come at somewhere between 230k and 270k. Time to talk to my realtor!

Friend: Glad I could help.

In the above conversation, we said that using our neighbor Alice's house as a reference is a good

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